

Moment thresholds for exact causal randomness

Information spectra, adaptive envelopes, and a logarithmic coefficient

26 September 2026

Abstract

Consider exact simulation of two finite output laws, selected adaptively for n rounds, using a single initial discrete seed. For equal-entropy pure dyadic laws, we prove that the minimum seed entropy is nh plus an excess of order $\sqrt{n}$ when information variances differ, of order $\log n$ when only the third centered information moments first differ, and bounded when both moments agree. The result follows from exact attainment of a recursive information-spectrum bound and a moment classification of controlled random-walk envelopes. The latter classification holds for arbitrary finite centered increment laws. The proof uses an exact L^1 entropy-increment identity, a Fourier lower bound, and an integrated envelope contraction with compact smoothing. We also formulate a sharp logarithmic-coefficient conjecture, isolate a sufficient L^1 remainder estimate, and report reproducible computations for seven dyadic families. The coefficient remains unproved.

Keywords. Exact simulation; minimum-entropy coupling; information spectrum; controlled random walk; moment matching.

1 Introduction

The entropy needed to supply independent outputs of one prescribed law is additive. A different question arises when the law can be selected after observing earlier outputs, while all randomness must be supplied at time zero. Equal output entropies remove the first-order distinction between the actions. They do not remove the cost of supporting both choices at every observable history.

This paper studies that excess cost in a history-indexed model. Its main result identifies a finite moment threshold: for pure dyadic action laws, matching information variance and the third centered information moment is necessary and sufficient for uniformly bounded excess. A third-moment defect produces a logarithm; a variance defect produces a square root. The theorem classifies every finite pure dyadic pair of equal entropy, without symmetry assumptions. It does not provide a uniform constant as the pair varies.

The proof separates two questions. First, a recursive spectral bound is exactly realizable by a seed in the pure dyadic class. Second, the mean of that spectral envelope can be classified analytically. The analytic part does not require a lattice or dyadic probabilities. This distinction also identifies the obstruction to extending the causal upper bounds: for general action laws, an envelope need not be the information spectrum of an exactly realizable seed.

Relation to existing work. Minimum-entropy coupling and functional representation provide the one-round setting. Majorization-based constructions and approximation bounds were developed by Cicalese, Gargano, and Vaccaro [1] and Li [2]. Information-spectrum converses for coupling and functional representation were established by Shkel and Yadav [3]; Compton et al. [4] developed a profile lower bound beyond the majorization bound. The spectral ordering used below belongs to this established framework. Our argument applies it recursively under conditional, history-by-history constraints.

The approximation landscape has also changed recently. Compton’s additive- ε scheme [5] is polynomial in support size for a fixed number of marginals and fixed ε . Minimum-Rényi entropy coupling [6] and the geometry of Rényi entropy on the majorization lattice [7] address related static questions. These results do not, as stated, give a horizon-uniform error bound for the repeated conditional construction considered here. The claims proved in this paper concern exact recursive attainment and the asymptotics of its adaptive envelope; they are not claims to originate majorization or information-spectrum lower bounds.

Martingale and nonlinear limit theory. There is a second relevant literature. The controlled sums below are martingales, and logarithmic losses in martingale normal approximation are well established; Guo [8] gives recent Wasserstein bounds and a detailed account of earlier results, including Bolthausen’s sharp Kolmogorov rate. Fang, Peng, Shao, and Song [9] obtain $(1 + \log n)/\sqrt{n}$ bounds for Lipschitz tests under sublinear expectations. Thus a logarithm arising from adaptive increments is not by itself an unfamiliar phenomenon. Our quantity is the integral of a pointwise envelope over policies. The optimizing policy can vary with the integration threshold, so a bound on the Wasserstein distance of each individual controlled law does not directly bound this quantity. The integrated contraction below deals with that order of optimization and integration explicitly.

Organization and status. Sections 2–8 give the classification and its proof. Section 9 gives an explicit logarithmic pair with all-horizon numerical constants. Section 10 separates a conjectured coefficient from a proved conditional criterion, and Section 11 reports finite computations. The manuscript makes no claim of established literature priority. In particular, the sharp coefficient is not included in the classification theorem.

2 Model and main results

All information quantities use base-two logarithms. For a finite law A , write $\iota_A(x) = -\log_2 A(x)$ on its positive support. Define

$$h_A = \mathbb{E}\iota_A(X), \quad v_A = \mathbb{E}(\iota_A(X) - h_A)^2, \quad \kappa_A = \mathbb{E}(\iota_A(X) - h_A)^3, \quad X \sim A.$$

A *pure dyadic law* has each positive atom equal to 2^{-k} for an integer $k \geq 0$. The restriction is to individual powers of two, not to arbitrary rational numbers with a power-of-two denominator.

A horizon- n simulator consists of one discrete seed S , independent of any policy randomization, and deterministic maps

$$Y_t = f_t(S, A_1, Y_1, \dots, A_{t-1}, Y_{t-1}, A_t), \quad 1 \leq t \leq n,$$

where $A_t \in \{P, Q\}$. A policy selects A_t using the observed history (and, if desired, independent private randomness). Admissibility means

$$\text{Law}(Y_t \mid A_1, Y_1, \dots, A_{t-1}, Y_{t-1}, A_t) = A_t$$

at every positive-probability conditioning event, for every such policy. There is no fresh simulator randomness after time zero. It suffices to check deterministic policies: conditioning on a policy’s private randomness gives the general case. The policy observes past actions and outputs, not the seed. Let $C_n^{\text{causal}}(P, Q)$ be the infimum seed entropy at horizon n . The seed is allowed to depend on n . No identification of responses at different histories is imposed merely because the same action appears there. All positive atom probabilities are included and zero-probability symbols are omitted. We use $\ln$ for the natural logarithm, including in the sharp coefficient below.

Theorem 1 (Complete equal-entropy classification for pure dyadic laws). *Let P, Q be finite pure dyadic laws with $h_P = h_Q = h$, and set $\Delta_n = C_n^{\text{causal}}(P, Q) - nh$. The optimum is attained by an explicit finite seed. The following cases are exhaustive:*

$v_P \neq v_Q$	$: \Delta_n = \Theta(\sqrt{n}),$
$v_P = v_Q, \kappa_P \neq \kappa_Q$	$: \Delta_n = \Theta(\log(n+1)),$
$v_P = v_Q, \kappa_P = \kappa_Q$	$: \Delta_n = O(1).$

In the bounded case, Δ_n is nondecreasing and converges to a finite limit. This limit is positive unless P, Q are permutations of one another; in that exceptional case $\Delta_n = 0$ for all n . All asymptotic constants may depend on the fixed pair.

Thus, within this class, the second and third centered information moments completely determine the growth order. Higher moments may affect the bounded limit but do not create further unbounded orders.

Corollary 2 (Necessary conditions without dyadic assumptions). *For arbitrary finite action laws of equal entropy, unequal information variances imply $\Delta_n = \Omega(\sqrt{n})$. Equal variances and unequal third centered information moments imply $\Delta_n = \Omega(\log(n+1))$. In particular, bounded overhead requires matching both moments.*

Theorem 3 (Moment classification of adaptive envelopes). *Let α, β be any finite centered probability laws on $\mathbb{R}$. Define*

$$G_0 = \mathbf{1}_{[0, \infty)}, \quad G_{n+1} = \min\{\alpha * G_n, \beta * G_n\},$$

and let e_n be the mean of the probability law with CDF G_n . Then e_n is nonnegative and nondecreasing, and

$$e_n = \begin{cases} \Theta(\sqrt{n}), & \int x^2 \alpha(dx) \neq \int x^2 \beta(dx), \\ \Theta(\log(n+1)), & \int x^2 \alpha(dx) = \int x^2 \beta(dx), \quad \int x^3 \alpha(dx) \neq \int x^3 \beta(dx), \\ O(1), & \int x^j \alpha(dx) = \int x^j \beta(dx), \quad j = 2, 3. \end{cases}$$

In the last case, the limiting mean is positive if $\alpha \neq \beta$, and zero otherwise. The theorem imposes no arithmetic restriction on supports or probabilities.

For general finite equal-entropy action laws, this theorem classifies a lower bound on causal seed entropy. Exact causal attainment is the additional dyadic statement in Lemma 4.

3 Exact profile recursion and dyadic realization

For any discrete law A , let $F_A(t) = \mathbb{P}\{\iota_A(X) \leq t\}$, and let $\mu_A = \text{Law}(\iota_A(X))$. Define CDFs recursively by

$$F_0(t) = \mathbf{1}_{\{t \geq 0\}}, \quad F_n(t) = \min\{(\mu_P * F_{n-1})(t), (\mu_Q * F_{n-1})(t)\}. \quad (3.1)$$

Convolution means $(\mu * F)(t) = \int F(t-x) \mu(dx)$. The minimum of two finite-support CDFs is again a finite-support CDF.

Lemma 4 (Profile lower bound and exact dyadic attainment). *For any finite P, Q , every admissible discrete horizon- n seed S satisfies $F_S(t) \leq F_n(t)$ for all real t . For pure dyadic P, Q , F_n is the information CDF of a constructible finite pure dyadic seed R_n and $C_n^{\text{causal}} = H(R_n)$.*

Proof. At horizon zero every seed has nonnegative information. Suppose the profile bound holds for $n - 1$ and fix the first action a with output Y . For every y of positive probability, $S \mid Y = y$ is an admissible seed for the remaining horizon. On that fiber,

$$-\log_2 \mathbb{P}\{S = s\} = \iota_a(y) - \log_2 \mathbb{P}\{S = s \mid Y = y\}.$$

Induction gives $F_S(t) \leq (\mu_a * F_{n-1})(t)$ for both first actions, proving the lower bound. Integrating survival functions bounds seed entropy below by the mean of F_n .

For finite pure dyadic laws R, A , a deterministic map from R to A exists if and only if $F_R(k) \leq F_A(k)$ for all integers k . Necessity follows because a source atom must fit inside its target atom. For sufficiency, assign source atoms in decreasing order of size. At mass 2^{-k} , all eligible remaining target capacities are integer multiples of 2^{-k} and total $F_A(k) - F_R(k - 1)$, enough for the current level. Thus all bins can be filled exactly.

For $F(k) = \min(F_A(k), F_B(k))$, the counts $N_k = 2^k(F(k) - F(k - 1))$ are nonnegative integers. Taking N_k atoms of mass 2^{-k} realizes F and refines both A, B . Apply this recursively to $P \otimes R_{n-1}$ and $Q \otimes R_{n-1}$, starting at a point mass R_0 . The refinement maps send the current state to an output and a new state. Conditional on that output, the new state has exactly law R_{n-1} . Induction proves correctness after every history and for every adaptive policy. The profile lower bound proves optimality, even against non-dyadic seeds. $\square$

4 The envelope and its exact entropy increments

Assume henceforth that both entropies equal h . Put $\alpha = \text{Law}(\iota_P - h)$ and $\beta = \text{Law}(\iota_Q - h)$. Both laws are centered. Let a controller choose either increment law predictably, and let S_j^π be its sum after j steps. Define

$$G_j(x) = \inf_{\pi} \mathbb{P}\{S_j^\pi \leq x\}, \quad U_j(x) = \sup_{\pi} \mathbb{P}\{S_j^\pi \leq x\}.$$

Dynamic programming, conditioning on the first increment, gives

$$G_j = \min(\alpha * G_{j-1}, \beta * G_{j-1}), \quad G_0 = \mathbf{1}_{[0, \infty)}.$$

The optimizing policy may depend on the threshold x ; the envelope need not be the CDF of any single controlled walk. Consequently $G_j(x) = F_j(jh + x)$. Let Z_j have CDF G_j and put $e_j = \mathbb{E}Z_j$. For all finite action laws $\Delta_j \geq e_j$; for pure dyadic laws, Lemma 4 gives equality. The two CDFs in the minimum have the same mean e_{j-1} . Therefore

$$e_{j+1} - e_j = \frac{1}{2} \|(\alpha - \beta) * G_j\|_1 \geq 0. \quad (4.1)$$

Indeed, use $\min(A, B) = (A + B - |A - B|)/2$ and the CDF formula for differences of means. This identity charges a nonnegative cost at every horizon.

Write $B = \max\{|x| : x \in \text{supp } \alpha \cup \text{supp } \beta\}$. If $B = 0$, both information laws coincide and the problem is trivial. Assume $B > 0$. For every policy, conditional centering and $|D_i| \leq B$ imply

$$\mathbb{P}\{S_j^\pi \geq t\}, \mathbb{P}\{S_j^\pi \leq -t\} \leq \exp\left(-\frac{t^2}{2jB^2}\right), \quad t > 0.$$

For completeness, convexity gives $\mathbb{E}[e^{sD_i} \mid \mathcal{F}_{i-1}] \leq \cosh(sB) \leq e^{s^2 B^2/2}$; iterate and optimize the exponential Markov bound. The estimate is uniform over policies, hence also bounds both tails of Z_j . Integrating the tails yields

$$\mathbb{E}Z_j^2 \leq 4B^2j, \quad 0 \leq e_j \leq 2B\sqrt{j}. \quad (4.2)$$

5 A Fourier lower bound: why a logarithm is unavoidable

Let $D = \alpha - \beta$, and use the Fourier convention $\widehat{\nu}(t) = \int e^{itx} \nu(dx)$. Suppose $d \geq 2$ is the first order at which the moments of α, β differ. Finite support and Taylor expansion imply that for some $c_0, t_0 > 0$,

$$|\widehat{D}(t)| \geq c_0 |t|^d \quad (0 < |t| \leq t_0). \quad (5.1)$$

Choose $c = \min(t_0, 1/(2B))$ and $t_j = c/\sqrt{j+1}$. Equation (4.2) and $\cos u \geq 1 - u^2/2$ show

$$|\widehat{\text{Law}(Z_j)}(t_j)| \geq \Re \widehat{\text{Law}(Z_j)}(t_j) \geq 1 - \frac{1}{2} t_j^2 \mathbb{E} Z_j^2 \geq \frac{1}{2}.$$

The function $R_j = D * G_j$ is compactly supported, and its distributional derivative is $D * \text{Law}(Z_j)$. Integration by parts therefore gives

$$\|R_j\|_1 \geq |\widehat{R}_j(t_j)| = \frac{|\widehat{D}(t_j)| |\widehat{\text{Law}(Z_j)}(t_j)|}{|t_j|} \geq \frac{c_0 c^{d-1}}{2} (j+1)^{-(d-1)/2}.$$

Together with (4.1), this proves

$$e_n \geq \frac{c_0 c^{d-1}}{4} \sum_{j=0}^{n-1} (j+1)^{-(d-1)/2}. \quad (5.2)$$

For $d = 2$ this is $\Omega(\sqrt{n})$; for $d = 3$ it is $\Omega(\log(n+1))$. Since $\Delta_n \geq e_n$ without a dyadic assumption, Corollary 2 follows already. The upper estimate in (4.2) proves the matching square-root upper bound for the envelope, and hence for the causal excess in the pure dyadic class.

6 An integrated envelope contraction

For the remaining upper bounds, put

$$W_j = \int_{\mathbb{R}} (U_j - G_j)(x) dx.$$

The all- β walk has mean zero and its CDF lies between the envelopes. Thus $0 \leq e_j \leq W_j$. No reflection symmetry is assumed here. To distinguish this width from an ordinary Wasserstein bound, for any reference CDF B_j lying between the envelopes one has

$$\sup_{\pi} \int |F_j^{\pi} - B_j| dx \leq \int \sup_{\pi} |F_j^{\pi} - B_j| dx \leq W_j.$$

In general the first inequality cannot be reversed. In particular, a lower bound on e_j does not establish the same lower bound on the Wasserstein error of one controlled walk.

Let V be independent smoothing noise supported in $[-A, A]$, let T_r be the all- β walk, and define

$$h_r(x) = \mathbb{P}\{T_r + V \leq x\}, \quad R_r = D * h_r, \quad K_r = |R_r|, \quad a_r = \|K_r\|_1, \quad b_r = \|K'_r\|_1.$$

The noise chosen below ensures that these functions are absolutely continuous and compactly supported where needed.

Lemma 5 (Contraction). *For every $n \geq 1$,*

$$W_n \leq 2A + 2 \sum_{r=0}^{n-1} (a_r + b_r W_{n-1-r}). \quad (6.1)$$

If $b_ := \sum_{r \geq 0} b_r < 1/2$, then for every $N \geq 1$,*

$$\max_{0 \leq n \leq N} W_n \leq \frac{2A + 2 \sum_{r=0}^{N-1} a_r}{1 - 2b_*}. \quad (6.2)$$

Proof. Write G_n^V, U_n^V for the smoothed policy envelopes and W_n^V for their integrated width. Bounded support of V gives $U_n(x) \leq U_n^V(x + A)$ and $G_n(x) \geq G_n^V(x - A)$. Every CDF F satisfies $\int (F(x + A) - F(x - A))dx = 2A$ by Tonelli, so $W_n \leq W_n^V + 2A$.

For each policy, telescope $\mathbb{E}h_{n-i}(x - S_i^\pi)$ over i . Since $h_{r+1} = \beta * h_r$, conditioning on the past gives

$$\mathbb{P}\{S_n^\pi + V \leq x\} - h_n(x) = \sum_{i=1}^n \mathbb{E} \left[\mathbf{1}_{\{\text{action } i \text{ uses } \alpha\}} R_{n-i}(x - S_{i-1}^\pi) \right].$$

Its absolute value is at most $\sum_i \mathbb{E}K_{n-i}(x - S_{i-1}^\pi)$. Let B_j be the CDF of T_j . For every controlled prefix CDF F_j^π , $|F_j^\pi - B_j| \leq U_j - G_j$. Stieltjes integration by parts gives

$$\mathbb{E}K_r(x - S_j^\pi) \leq \mathbb{E}K_r(x - T_j) + \int |K_r'(x - y)|(U_j(y) - G_j(y)) dy.$$

The right side is independent of π , so taking the supremum first and then integrating in x proves

$$\int \sup_{\pi} \mathbb{E}K_r(x - S_j^\pi) dx \leq a_r + b_r W_j.$$

The smoothed envelope width is at most twice the largest absolute deviation from h_n at each threshold. Summing proves (6.1). Taking a maximum over $n \leq N$, with $W_0 = 0$, and moving $2b_*$ times that finite maximum to the left proves (6.2). $\square$

7 Derivative estimates for an arbitrary finite reference law

Assume β is nondegenerate. Choose two of its support points $x < y$, put $b = (y - x)/2 > 0$ and $\rho = 2 \min(\beta\{x\}, \beta\{y\}) > 0$. Then $\beta = \rho\eta + (1 - \rho)\zeta$, where $\eta = (\delta_x + \delta_y)/2$ and ζ is a probability law when $\rho < 1$. The case $\rho = 1$ needs no residual law. Let v_a denote the uniform density on $[-a, a]$, choose $a = Lb$ with a positive integer L , and take V to be the sum of four independent variables with density v_a . Then $A = 4a$ and, in weak-derivative notation,

$$h_r^{(q+1)} = (v'_a)^{*q} * v_a^{*(4-q)} * \beta^{*r}, \quad q = 2, 3, 4. \quad (7.1)$$

Here $v'_a = (\delta_{-a} - \delta_a)/(2a)$, and exponent zero means δ_0 . In particular $\|h_r^{(q+1)}\|_{\text{TV}} \leq a^{-q}$. The fourth case can be a signed measure; for densities the total variation norm agrees with the L^1 norm.

Lemma 6. *There are constants $C_q < \infty$ depending on β, q , but independent of $a = Lb$ and r , such that*

$$\|h_r^{(q+1)}\|_{\text{TV}} \leq \min\{a^{-q}, C_q(r + 1)^{-q/2}\}, \quad q = 2, 3, 4. \quad (7.2)$$

Proof. Telescoping point masses shows $v'_a = \nu_L * v'_b$, where ν_L is uniform on $\{-a + b, -a + 3b, \dots, a - b\}$. Convolution with probability measures cannot increase total variation. In the binomial mixture expansion of β^{*r} , the number K of η increments has law $\text{Bin}(r, \rho)$. For a fixed $K = k$, translation and scaling of symmetric signs give

$$\|(v'_b)^{*q} * \eta^{*k}\|_{\text{TV}} \leq b^{-q} \binom{k+q}{q}^{-1/2}.$$

To verify this inequality, let $N = k + q$ independent symmetric signs be ε_i . Up to sign and the factor b^{-q} , the signed measure is the law of their sum weighted by $\prod_{i=1}^q \varepsilon_i$. Given the sum, the conditional product equals the average of $\prod_{i \in I} \varepsilon_i$ over all q -element subsets I . These products are orthonormal, so that average has squared L^2 norm $\binom{N}{q}^{-1}$. Cauchy–Schwarz proves the asserted variation bound.

It remains to average over K . Using $\binom{k+q}{q} \geq (k+1)^q/q!$ and splitting at $K = \rho r/2$ gives

$$\mathbb{E} \binom{K+q}{q}^{-1/2} \leq \sqrt{q!} (1 + \rho r/2)^{-q/2} + e^{-\rho r/8} \leq C'_q (r+1)^{-q/2}.$$

For the exponential term, apply Markov to $e^{-(\ln 2)K}$ and use $1 - \rho/2 \leq e^{-\rho/2}$ and $\ln 2 < 3/4$. This proves the decay estimate; combine it with the direct bound a^{-q} from (7.1). $\square$

8 Closing the classification

The remaining step is a cancellation estimate that does not use a lattice. Suppose the moments of $D = \alpha - \beta$ vanish through order $m-1$. Taylor's formula with integral remainder gives

$$D * f = \Gamma_m * f^{(m)}, \quad \Gamma_m = \int \frac{(-y)^m}{(m-1)!} \int_0^1 (1-t)^{m-1} \delta_{ty} dt D(dy). \quad (8.1)$$

The terms of degrees below m vanish after integration against D . The signed measure Γ_m is finite and compactly supported, with

$$\|\Gamma_m\|_{\text{TV}} \leq M_m := \frac{1}{m!} \int |y|^m |D|(dy).$$

The identity first holds for smooth f ; convolution with a smooth approximate identity extends it to the weak derivatives used here. For the smoothing of Lemma 6, $R_r = D * h_r$ is absolutely continuous and its derivative is integrable. Since $\|(|R_r|)'\|_1 \leq \|R'_r\|_1$, we obtain

$$a_r \leq M_m \|h_r^{(m)}\|_{\text{TV}}, \quad b_r \leq M_m \|h_r^{(m+1)}\|_{\text{TV}}. \quad (8.2)$$

If the common variance is positive, the reference law is nondegenerate and Lemma 6 applies. For $m = 3$ or 4 , the summable sequence $M_m C_m (r+1)^{-m/2}$ dominates the bounds on b_r . As $a = Lb \rightarrow \infty$, their pointwise upper bound $M_m a^{-m}$ tends to zero. Dominated convergence lets us fix a finite a with $\sum b_r < 1/2$.

If variances match but third moments differ, use $m = 3$. Then $a_r \leq M_3 C_2 / (r+1)$, and (6.2) gives $W_n = O(\log(n+1))$. Since $e_n \leq W_n$, the Fourier lower bound proves the logarithmic case of Theorem 3.

If variances and third moments both match, use $m = 4$. Then $a_r \leq M_4 C_3 (r+1)^{-3/2}$ is summable, so (6.2) gives $W_n = O(1)$ and $e_n = O(1)$. If the common variance is zero, both centered increment laws are δ_0 and this conclusion holds trivially. Finally (4.1) proves monotonicity. When $\alpha \neq \beta$, it also gives $e_1 = \frac{1}{2} \int |F_\alpha - F_\beta| > 0$. This completes Theorem 3. For pure dyadic action laws, $\Delta_n = e_n$ by Lemma 4. Equality of the information laws is equivalent to equality of all atom multiplicities, hence to being permutations of one another. In that case the recursion is the ordinary product profile and $\Delta_n = 0$. This completes Theorem 1.

9 An explicit logarithmic pair

Take

$$P = (1/16, (1/32)^{\times 10}, (1/64)^{\times 12}, (1/128)^{\times 56}), \\ Q = ((1/32)^{\times 16}, (1/128)^{\times 64}).$$

Their support sizes are 79 and 80. Their centered information laws are

$$\alpha = \frac{1}{16} \delta_{-2} + \frac{5}{16} \delta_{-1} + \frac{3}{16} \delta_0 + \frac{7}{16} \delta_1, \quad \beta = \frac{1}{2} \delta_{-1} + \frac{1}{2} \delta_1.$$

Direct calculation gives

$$h_P = h_Q = 6, \quad v_P = v_Q = 1, \quad \kappa_P = -\frac{3}{8}, \quad \kappa_Q = 0, \quad D = \frac{1}{16}\delta_{-2} * (\delta_0 - \delta_1)^{*3}.$$

Let $H_n = \sum_{k=1}^n 1/k$ be the harmonic number.

Theorem 7 (A genuine logarithmic overhead). *For the displayed fixed pair and every $n \geq 1$,*

$$\boxed{\frac{H_n}{2048} \leq C_n^{\text{causal}}(P, Q) - 6n < \frac{128}{5} + \frac{3}{10}H_n.} \quad (9.1)$$

In particular, the overhead is $\Theta(\log n)$ and is unbounded. The exact one-step excess is $\Delta_1 = 1/8$.

Proof. For the lower bound, $B = 2$ in (4.2). Choose $t_j = 1/(4\sqrt{j+1})$, so $|\widehat{\text{Law}}(\widehat{Z}_j)(t_j)| \geq 1/2$. For $0 \leq t \leq 1/4$, the inequality $2 \sin(t/2) \geq t - t^3/24$ implies $|1 - e^{it}|^3 \geq t^3/2$. Thus $|\widehat{D}(t_j)| \geq t_j^3/32$, and (4.1) gives

$$e_{j+1} - e_j \geq \frac{1}{2} \frac{t_j^3/32}{t_j} \cdot \frac{1}{2} = \frac{1}{2048(j+1)}.$$

Summation proves the lower bound.

For the upper bound use four uniform smoothing variables on $[-2, 2]$, so $a = 2$ and $A = 8$. For the Rademacher reference, the derivative estimate has the explicit form

$$\|h_r^{(q+1)}\|_{\text{TV}} \leq \min\{2^{-q}, \binom{r+q}{q}^{-1/2}\}.$$

Here the cubic finite-difference factorization of D is sharper than the general Taylor bound: the third unit difference is the third weak derivative convolved, up to sign, with $\mathbf{1}_{[0,1]}^{*3}$. Its convolution factor has L^1 norm one. Thus $a_r \leq \|h_r^{(3)}\|_{\text{TV}}/16$ and $b_r \leq \|h_r^{(4)}\|_{\text{TV}}/16$, giving

$$\sum_{r=0}^{n-1} a_r \leq \frac{\sqrt{2}}{16} H_n, \quad b_* \leq \frac{1}{16} \sum_{r \geq 0} \min\{2^{-3}, \sqrt{6}(r+1)^{-3/2}\} \leq \frac{3\sqrt[3]{6}}{32}.$$

For the last sum, bound it by $\int_0^\infty \min(2^{-3}, \sqrt{6}x^{-3/2})dx = 3\sqrt[3]{6}/2$. Therefore (6.2) and $e_n \leq W_n$ give

$$\Delta_n \leq \frac{16 + \frac{\sqrt{2}}{8}H_n}{1 - \frac{3\sqrt[3]{6}}{16}} < \frac{128}{5} + \frac{3}{10}H_n,$$

using $\sqrt{2} < 3/2$ and $\sqrt[3]{6} < 2$. Finally the CDF of $\alpha - \beta$ is $1/16, -2/16, 1/16$ on the unit intervals starting at $-2, -1, 0$. Equation (4.1) gives $\Delta_1 = (1 + 2 + 1)/32 = 1/8$. $\square$

10 A conjectured universal logarithmic coefficient

The classification determines growth order, not the leading constant. The increment identity suggests a stronger law. Write $\delta\kappa = \int x^3 D(dx)$ and suppose the common variance is $v > 0$. Let Φ and ϕ be the standard normal CDF and density, and set

$$\Psi_n(x) = \Phi\left(\frac{x}{\sqrt{nv}}\right), \quad c(\alpha, \beta) = \frac{|\delta\kappa|}{3v\sqrt{2\pi e}}.$$

Conjecture 8 (Sharp logarithmic coefficient). *For finite pure dyadic action laws with common entropy h , common information variance $v > 0$, and $\kappa_P \neq \kappa_Q$,*

$$C_n^{\text{causal}}(P, Q) = nh + \frac{|\kappa_P - \kappa_Q|}{3v\sqrt{2\pi e}} \ln n + O(1). \quad (10.1)$$

The remainder may depend on the pair.

The coefficient is in bits per unit of $\ln n$. If the logarithm in (10.1) is written as $\log_2 n$, the displayed coefficient must be multiplied by $\ln 2$.

Formal origin of the constant. Since the first three moment cancellations include orders 0, 1, 2, Taylor expansion against a smooth profile gives

$$D * \Psi_n = -\frac{\delta\kappa}{6}\Psi_n''' + \text{higher-order terms.}$$

Moreover,

$$\Psi_n'''(x) = \frac{1}{(nv)^{3/2}} \left(\frac{x^2}{nv} - 1 \right) \phi\left(\frac{x}{\sqrt{nv}}\right), \quad \|\Psi_n'''\|_1 = \frac{4\phi(1)}{nv}.$$

For the last equality, integrate $(1 - z^2)\phi(z)$ on $[-1, 1]$, using $(z\phi(z))' = (1 - z^2)\phi(z)$, and use the zero integral of $(z^2 - 1)\phi(z)$ on $\mathbb{R}$. Substitution into the exact increment identity predicts $e_{n+1} - e_n \sim c(\alpha, \beta)/n$. The missing step is control of the actual nonlinear envelope at the scale of this differentiated L^1 expression. A central limit theorem for the undifferentiated CDF alone would not justify that substitution.

Proposition 9 (A sufficient remainder estimate). *Let α, β be finite centered laws of common variance $v > 0$, with $\delta\kappa \neq 0$. Define*

$$\varepsilon_n = \left\| D * G_n + \frac{\delta\kappa}{6}\Psi_n''' \right\|_1, \quad n \geq 1.$$

Then

$$\left| (e_{n+1} - e_n) - \frac{c(\alpha, \beta)}{n} \right| \leq \frac{\varepsilon_n}{2}. \quad (10.2)$$

If $\varepsilon_n = o(n^{-1})$, then $e_n = c(\alpha, \beta) \ln n + o(\ln n)$. If $\sum_{n \geq 1} \varepsilon_n < \infty$, then $e_n - c(\alpha, \beta) \ln n$ converges to a finite limit.

Proof. The reverse triangle inequality, (4.1), and the computed norm of Ψ_n''' give (10.2). Sum over n . An $o(1/n)$ sequence has partial sums $o(\ln n)$; absolute summability gives convergence after subtracting the harmonic sum. Finally $H_n - \ln n$ converges. $\square$

The proposition is conditional: neither of its remainder hypotheses is proved here for the full class in Conjecture 8. The summable estimate is stronger than what is needed for an $O(1)$ remainder, but it specifies one concrete sufficient analytic target.

Consistency checks. Several transformations constrain a proposed coefficient. Reflection changes the sign of $\delta\kappa$ but not the growth cost. Dilation of both centered increment laws by $a > 0$ gives exactly $G_n^{(a)}(x) = G_n(x/a)$ and $e_n^{(a)} = ae_n$; the proposed coefficient has the same scaling since $v \mapsto a^2v$ and $\delta\kappa \mapsto a^3\delta\kappa$. Adding the same independent centered noise to each increment leaves $\delta\kappa$ unchanged and raises v . The formula therefore predicts a smaller logarithmic coefficient. This last prediction is a consequence of the conjecture, not an additional theorem.

11 Numerical evidence and exact finite certificates

We evaluate the CDF recursion deterministically on its complete integer grid; there is no sampling, fitted coefficient, or tail truncation. Let α, β be the centered laws in Section 9, and put

$$\gamma = \frac{1}{8}\delta_{-2} + \frac{3}{4}\delta_0 + \frac{1}{8}\delta_2, \quad \check{\alpha} = \text{Law}(-X), \quad X \sim \alpha.$$

In Table 1, 2α denotes the pushforward under $x \mapsto 2x$, not multiplication of total mass. All listed laws are centered with dyadic rational weights. Each pair therefore admits an exact realization as centered information laws of finite pure dyadic action laws: if all weights have denominator dividing 2^s and the support lies in $[-K, K] \cap \mathbb{Z}$, choose $h = s + K$ and replace mass p_k at k by

$p_k 2^{h+k}$ atoms of probability $2^{-(h+k)}$. These are integer multiplicities, the total mass is one, and the entropy is h .

Increment laws	v	$ \delta\kappa $	c	$n(e_n - e_{n-1})$	$10^4(r_n - 1)$
α, β	1	0.375	0.03024634	0.03025164	1.752
$(\alpha + \beta)/2, \beta$	1	0.1875	0.01512317	0.01512696	2.506
α, γ	1	0.375	0.03024634	0.03025556	3.047
$\alpha, \tilde{\alpha}$	1	0.75	0.06049268	0.06049348	0.132
$\alpha * \beta, \beta * \beta$	2	0.375	0.01512317	0.01512514	1.305
$(\alpha + 3\beta)/4, (3\alpha + \beta)/4$	1	0.1875	0.01512317	0.01512413	0.632
$2\alpha, 2\beta$	4	3	0.06049268	0.06050328	1.752

Table 1: Deterministic floating-point recursion at $n = 10\,000$. Here $c = |\delta\kappa|/(3v\sqrt{2\pi e})$ is specified in advance and $r_n = n(e_n - e_{n-1})/c$. The last row is an exact scaling consistency check, not an independent family. Finite agreement does not establish the asymptotic formula.

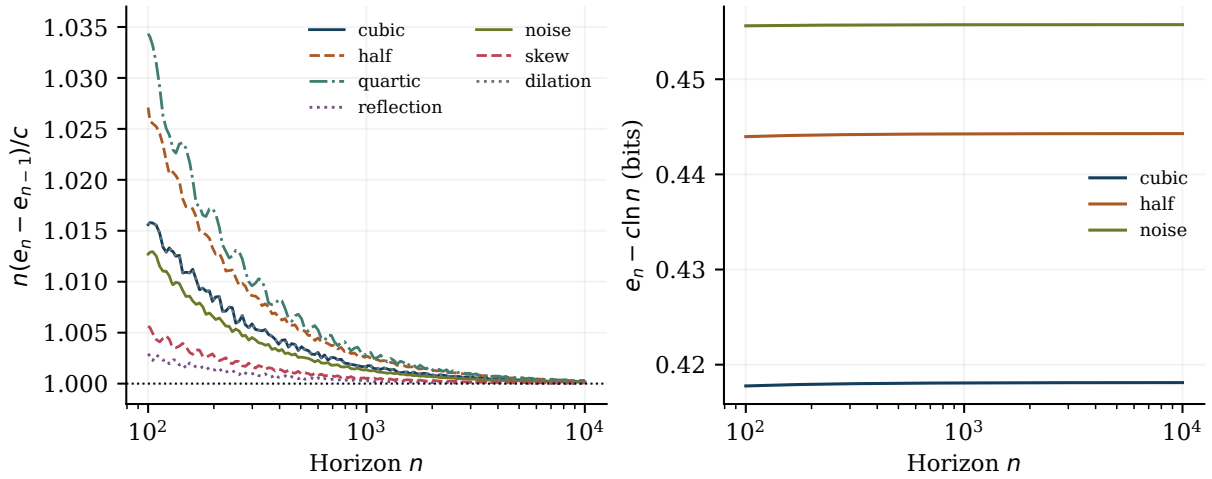


Figure 1: Left: normalized entropy increments approach the predicted coefficient over the displayed range. The cubic and dilation curves coincide by exact scaling. Right: centered excess for three selected families. Apparent stabilization over a finite range does not prove a bounded remainder. Full parameters and numerical rows are supplied in `numerical_results.json` and `numerical_results.csv`.

For each of the seven pairs, integer arithmetic verifies normalization, nonnegative profile masses, integer dyadic seed multiplicities, and the entropy-increment identity through horizon 128. Floating-point values at that horizon are compared with the exact rational result. For the cubic pair, a second run at $n = 10\,000$ using the platform's extended-precision `longdouble` changes $n(e_n - e_{n-1})$ by approximately 2.1×10^{-11} and e_n by approximately 1.1×10^{-11} . These comparisons check arithmetic sensitivity; they do not bound the asymptotic remainder.

A separate exact program constructs the full dyadic seed partitions through horizon two for the explicit action pair, and verifies the conditional product law of every decoder branch. Its seed sizes are 84 and 6984. At longer horizons it keeps atom multiplicities instead of expanding all seed states. In particular, it verifies

$$\Delta_1 = \frac{1}{8}, \quad \Delta_2 = \frac{7}{32},$$

and the inequalities of Theorem 7 at each of the 128 checked horizons. The all-horizon claims follow from the proofs above, not from these finite certificates.

12 Scope and remaining questions

The analytic envelope classification and the exact pure dyadic causal classification are proved separately. Outside the pure dyadic class, the present argument supplies necessary moment conditions for bounded causal excess but no general matching seed construction. Even when the analytic envelope has bounded mean, realization may introduce additional cost. This realization question is distinct from the sharp-coefficient question, which already arises within the class where attainment is exact.

The logarithmic regime is critical for the method: cancellation through order two produces an r^{-1} smoothing cost, while one further moment cancellation makes that cost summable. The Fourier argument proves that a genuine third-moment defect cannot be removed by a different seed construction in the dyadic class. Higher moments can change the bounded limit, but cannot create another divergent order in that class.

The numerical study tests a specific coefficient and its transformation rules. A proof would require finer control of the nonsmooth nonlinear recursion than the integrated contraction used for growth orders. Proposition 9 states one such target explicitly. No conclusion about a common infinite source is implied: a different finite seed is allowed at each horizon, and the expanded decoders can have exponential size.

A Constructing and checking a finite decoder

For a pure dyadic information profile F_n , let

$$N_{n,k} = 2^k (F_n(k) - F_n(k-1)).$$

Create $N_{n,k}$ labeled seed atoms of mass 2^{-k} and sort them from largest to smallest. For action a , the target bins are labeled pairs (y, z) , with capacities $a(y)R_{n-1}(z)$. Fill these bins with the sorted source atoms by the packing argument of Lemma 4. A seed atom's bin label determines both the output y and the next state z . The exact identity

$$\mathbb{P}\{Y = y, Z = z \mid A = a\} = a(y)R_{n-1}(z)$$

proves that the next state has the required law conditional on the observed output. Iterating gives the history-indexed simulator.

For the explicit pair, the certificate stores every source atom's information length and its target-bin index for both actions at horizons one and two. If the previous seed has M atoms and the target index is t , the output index and next-state index are respectively $\lfloor t/M \rfloor$ and $t \bmod M$. All partition checks use integer masses on a common dyadic grid.

For fixed finite dyadic action laws, the information support has width $O(n)$. Thus the profile can be updated in $O(n)$ arithmetic operations per horizon, using $O(n)$ stored profile values, without expanding the seed. These are arithmetic-operation counts: exact numerator bit lengths also grow with n . No polynomial bound on the expanded decoder size is asserted.

B Reproduction

The accompanying package includes the manuscript source, figure, exact finite certificates, complete numerical tables, and two Python programs. From the package directory, run

```
python3 experiments.py --n 10000 --exact-n 128
python3 logarithmic_seed.py --exact-n 128 --float-n 0
pdflatex -interaction=nonstopmode -halt-on-error moment_thresholds.tex
pdflatex -interaction=nonstopmode -halt-on-error moment_thresholds.tex
```

The first program uses NumPy and Matplotlib; the second uses only the Python standard library when `-float-n 0` is selected. The JSON report records Python and NumPy versions and the available extended precision. All probability parameters in the exact computations are rational numbers. The predicted coefficient is computed directly from those parameters, without regression.

References

- [1] F. Cicalese, L. Gargano, and U. Vaccaro. Minimum-entropy couplings and their applications. *IEEE Transactions on Information Theory*, 65:3436–3451, 2019. doi:10.1109/TIT.2019.2894519.
- [2] C. T. Li. Efficient approximate minimum entropy coupling of multiple probability distributions. *IEEE Transactions on Information Theory*, 67(8):5259–5268, 2021. doi:10.1109/TIT.2021.3076986.
- [3] Y. Y. Shkel and A. K. Yadav. Information spectrum converse for minimum entropy couplings and functional representations. *IEEE International Symposium on Information Theory*, 2023. arXiv:2305.05745.
- [4] S. Compton, D. Katz, B. Qi, K. Greenewald, and M. Kocaoglu. Minimum-entropy coupling approximation guarantees beyond the majorization barrier. *Proceedings of Machine Learning Research*, 206:10445–10469, 2023. PMLR paper and supplementary material.
- [5] S. Compton. Efficient ϵ -approximate minimum-entropy couplings. arXiv:2509.19598v3, 18 June 2026.
- [6] Y.-J. Ma, F. Wang, and X.-Y. Wu. Efficient approximate Minimum-Rényi entropy couplings. *Discrete and Continuous Dynamical Systems-S*, 18(10):2924–2936, 2025. doi:10.3934/dcdss.2025098.
- [7] A. K. Yadav and Y. Y. Shkel. Geometry of Rényi entropy on the majorization lattice. arXiv:2605.09655v2, 22 May 2026.
- [8] X. Guo. On the rate of convergence of the martingale central limit theorem in Wasserstein distances. arXiv:2407.16980, 2024.
- [9] X. Fang, S. Peng, Q.-M. Shao, and Y. Song. Limit theorems with rate of convergence under sublinear expectations. arXiv:1711.10649v2, 2018.